HIRaD: Hidden Interaction Inference from Predictive Radar Dynamics

Jaekwon Lee
Hansung University
Seoul, South Korea
dlwowlb@hansung.ac.kr

Jaekoo Lee*
Kookmin University
Seoul, South Korea
jakoo@kookmin.ac.kr

Abstract

Millimeter-wave radar point clouds provide a visually privacy-preserving representation for human motion analysis, but object-related interactions remain difficult to describe because the object evidence that defines the interaction is not always separable or consistently observable in sparse radar returns. We address this problem without direct object detection. Instead, we infer hidden interaction semantics from the temporal structure of observation-conditioned corrections to predicted body motion, motivated by the view that object affordances shape how the body moves. To this end, we propose HIRaD, a radar semantic sensing framework that organizes sparse radar returns into a predictive node-structured latent state of local motion carriers. It represents interaction-relevant cues as corrective residual trajectories between a motion-continuation prior and an observation-conditioned posterior, and compresses these trajectories into a language-aligned semantic bottleneck. A structured prefix conditions a frozen autoregressive language model on both the semantic bottleneck and node-level motion summaries, separating compact interaction-level cues from node-level motion evidence. Experiments show that our proposed method improves radar captioning and interaction understanding under weak object observability over strong baselines, while ablations validate the importance of radar-conditioned prefixing, latent state representation, and residual trajectory modeling. Our code is available at https://dlwowlb.github.io/HIRaD.

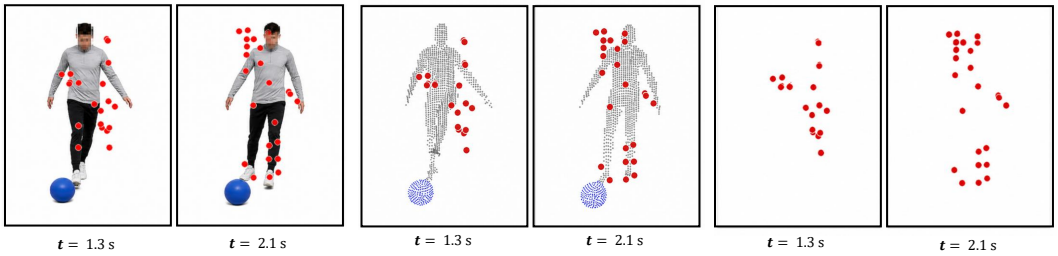
1 Introduction

Millimeter-wave radar captures physically grounded cues, including range, angle, Doppler velocity, and reflectivity, which can be represented as range-Doppler maps, range-angle heatmaps, or radar point clouds [1, 2]. Radar point clouds are particularly attractive for human sensing because they provide a compact 3D representation of the scene while retaining motion cues such as Doppler velocity [3]. However, as shown in Figure 1, this representation often favors strong and moving returns, making static or weakly reflective objects less reliable [4, 5]. These objects may be weakened by background filtering, sparsification, or unfavorable reflection properties, and can therefore appear fragmented, intermittent, or entirely absent [6]. Consequently, radar point clouds tend to preserve body motion more reliably than the object-level semantics that define the interaction.

This limitation is especially important for understanding object-involving human behavior from radar [7]. Unlike action classification [8, 9], which often maps motion patterns to a fixed set of labels, interaction understanding requires inferring the semantic cause of the motion. A model may observe that a person bends, reaches, kicks, or lowers the body, but still fail to infer the object-centered interaction behind the motion. Bending may indicate picking up an object, kicking may involve a ball, and lowering the body may correspond to sitting on a chair. Thus, unreliable object evidence

*Corresponding author

40th Conference on Neural Information Processing Systems (NeurIPS 2026).



(a) Vision-based Sensing (b) Lidar-based Sensing (c) Radar-based Sensing

Figure 1: Object observability in human–object interaction under different sensing modalities. (a) Vision- and (b) Lidar-based sensing provide relatively reliable observations of the interacted object, whereas (c) mmWave Radar-based sensing often preserves human motion but provides unreliable object evidence due to sparse and intermittent returns. The point-cloud visualizations in (a) and (b) are included only as auxiliary references for comparison.

creates a semantic gap: the body motion is observed, but the object-centered behavior that explains it may remain ambiguous.

We approach this gap from a predictive dynamics perspective. This view is motivated by two related observations. First, Gibson's notion of affordance [10] characterizes objects and environments in terms of the actions they make possible for an agent. Even when an object is not directly observed by radar, the action it affords can still shape how the body moves. Second, predictive coding [11] provides a useful perspective on perception as the relation between expected signals and observation-driven updates. Together, these motivate looking for hidden interaction cues not in the missing object itself, but in how object-induced constraints reshape the predictability and temporal structure of body-centric motion.

This motivates HIRaD, our proposed interaction-aware radar representation framework that models constraint-modulated predictive dynamics from sparse radar point clouds. It organizes sparse radar point clouds into a predictive node-structured latent state composed of temporally persistent local motion nodes. These nodes are not supervised anatomical landmarks; instead, they provide a stable body-centric substrate on which local motion continuation can be predicted. Given this state, the model forms a motion-continuation prior and an observation-conditioned posterior derived from the current radar evidence. Their regularized relationship yields residual-derived features that encode node-wise predictive update patterns.

It also encodes these residual-derived features as a structured representation of how local body dynamics are modulated by object- or surface-induced constraints. The node-wise temporal update pattern is preserved and summarized through a semantic bottleneck, producing a compact interaction-level cue. To express this cue as a language-based behavior description, our proposed method combines the semantic bottleneck with node-level motion evidence through a structured output interface.

Experiments on interaction-centric radar description show that our proposed method produces more accurate and semantically faithful descriptions than strong radar-based baselines. Ablation studies further show that radar-conditioned prefixing, latent state representation, and residual trajectory modeling each contribute to performance.

In summary, this paper makes three contributions: (1) we formulate hidden interaction inference in radar as a constraint-modulated predictive dynamics problem, where object-related cues are inferred from changes in the predictability and temporal structure of body-centric motion; (2) we introduce a predictive node-structured latent state that provides temporally persistent local motion nodes for modeling node-wise predictive update patterns; and (3) we propose a semantic bottleneck and structured output interface that combine compact residual-derived cues with node-level motion evidence for language-based interaction description.

2 Related work

Expanding frontiers of mmWave sensing. mmWave radar has rapidly expanded from a geometric ranging modality into a versatile perception medium. One line targets embodiment recovery, estimating skeletons or meshes from sparse point clouds (mm-Pose [12], mmMesh [13], P4Transformer [14], mRI [15]). A second line, neural radar reconstruction, ports neural implicit representations to RF: NeRF^2 [16] and Radar Fields [17] model wireless channels at scene scale, and GeRaF [18] reconstructs millimeter-level object geometry from raw RF through physics-based volumetric rendering. A third recognizes human activities from radar point clouds via supervised classification (RadHAR [1], Pantomime [19], MiliPoint [3], MMFi [20]). Together these recast mmWave from a geometric sensor into a versatile perception modality. Yet object recovery typically requires dense, near-range, multi-view scans of isolated objects, while activity classification remains limited to closed label sets. Interaction-level understanding from sparse single-view radar, where a moving human may interact with an unobserved object, remains largely unexplored.

Human–object interaction, affordance, and structured dynamics. Reasoning about what a body is doing with what draws on three converging lines. Vision-based HOI and affordance learning [21, 22, 23] predict interactions from visible human–object pairs, an assumption that breaks down for sparse single-view mmWave. Recent 3D-HOI generation instead models interaction as dynamics: HOI-Dyn [24] casts human–object motion as a driver–responder system, and HO-Flow [25] encodes hand–object trajectories on a unified latent manifold, both treating interaction as a coupled dynamical signature on the body. In parallel, slot-based and recurrent state-space models [26, 27, 28] provide the methodological substrate of structured latent state with temporal transitions. DAP [29] also uses pose prediction to learn temporal features for multi-camera 3D object detection. Our work connects these threads by using a structured predictive latent state for radar-observed body motion and reading interaction cues from posterior–prior corrections, rather than from directly observed object evidence.

Motion generation and captioning. Recent text-to-motion models, including MoMask++ [30], MoScale [31], and ParTY [32], offer alternative approaches to motion synthesis for dataset construction. A growing body of work translates motion into language. Skeleton-conditioned models (MotionGPT [33], TM2T [34], MotionLLM [35]) encode pose or motion sequences for pretrained language models. RadarLLM [7] is the closest radar-domain precedent: an Aggregate VQ-VAE-based motion-guided tokenizer compresses radar point-cloud sequences into discrete semantic tokens, which are aligned with a language model for radar-to-text generation. We follow the radar-token-conditioned LLM paradigm but augment it with a corrective residual derived interaction token that surfaces affordance information the radar return alone does not visibly encode.

3 Method

Overview. Given a sequence of sparse mmWave radar point clouds $\{\mathbf{P}_t\}_{t=1}^{T}$, our goal is to infer object-related interaction semantics from body-motion evidence and express them as a natural-language description. The central challenge is that the object defining the interaction is often weakly reflected, fragmented across frames, or not directly detected in the radar observation. Rather than attempting to detect or reconstruct the object itself, HIRaD uses a predictive body-centric representation to separate expected motion continuation from observation-conditioned updates. The method proceeds in three stages. First, it organizes sparse radar returns into a predictive node-structured latent state, whose nodes serve as temporally persistent local motion carriers. Second, it computes observation-supported corrective residuals between a motion-continuation prior and an observation-conditioned posterior. Third, it encodes node-wise residual histories with a deviation dynamics encoder, selects informative nodes using confidence and residual statistics, and compresses the result into a compact semantic token for caption generation.

3.1 Predictive Node-Structured Latent State

Before hidden interaction can be inferred, radar motion must be represented in a space where regular continuation is predictable. This is particularly important for mmWave point clouds, whose frame-wise observations are sparse, noisy, and temporally unstable. We therefore introduce M latent nodes $\{\mathbf{h}_t^{(m)}\}_{m=1}^{M}$ as local motion carriers. These nodes are not supervised anatomical landmarks. Instead,

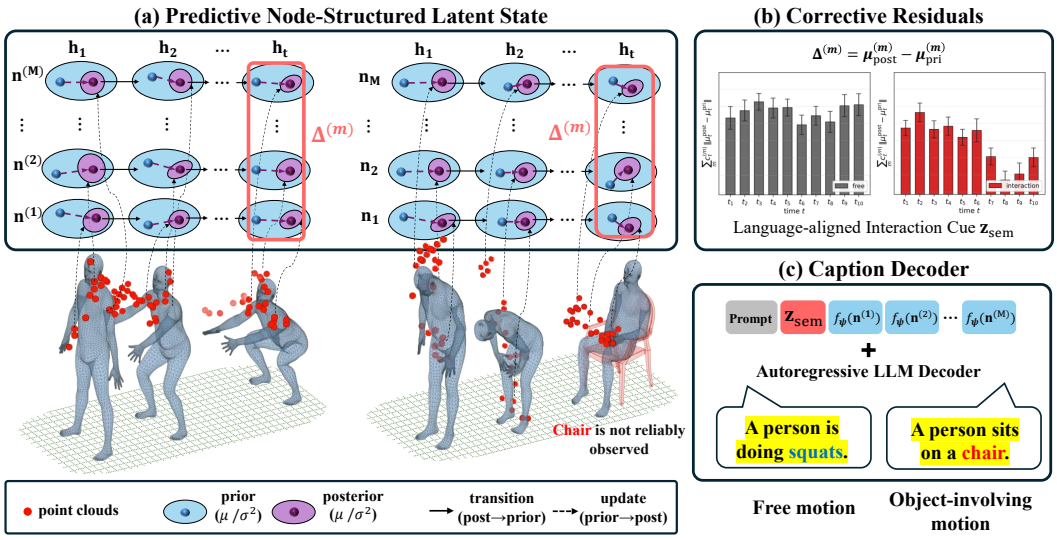


Figure 2: **Overview of the HIRaD framework.** Sparse radar point clouds are encoded into node-structured latent states. A motion prior is corrected by current observations to produce residual trajectories, which are converted into semantic and motion tokens for caption generation with a frozen language model.

they are latent nodes encouraged by routing and temporal persistence to capture local motion patterns over time.

Latent node assignment. At each frame, each valid radar point, represented by its 3D location and Doppler velocity, is mapped by a shared per-point MLP encoder f_{obs}. The resulting point features are routed to latent nodes by masked multi-head routing attention. This routing uses both node-wise and point-wise masked normalizations, encouraging competition among nodes and coherent point subsets per node (see Appendix F.1 for details). The routed features produce node-level evidence $\mathbf{e}_t^{(m)}$:

$$\mathbf{e}_t^{(m)} = W_o \underset{h=1}{\overset{H}{\text{Concat}}} \left(\sum_{n=1}^{N} a_t^{(m,n,h)} W_{v,h} f_{\text{obs}}(\mathbf{p}_t^{(n)}) \right). \tag{1}$$

Here $a_t^{(m,n,h)}$ is the masked soft assignment from point n to node m at attention head h. Conceptually, $\mathbf{e}_t^{(m)}$ is the routed observation content for node m: points with larger routing weights contribute more strongly to its evidence.

The same pre-softmax routing logits are also summarized into a per-node update gate $c_t^{(m)} \in [0,1]$ using a small network g_ϕ applied to logit statistics and the valid-point ratio. While $\mathbf{e}_t^{(m)}$ provides the routed observation content, $c_t^{(m)}$ controls how strongly this evidence updates node m relative to the predictive prior. The gate is not a contact label or a simple assignment count, but a routing-derived confidence for the posterior update (see Appendix F.2 for details).

Predictive prior and posterior update. To make the node-structured latent state predictive, a temporal transition model first estimates how the node states should evolve before seeing the current observation. This transition uses a short-horizon temporal context $\boldsymbol{\xi}_t$ computed only from past posterior node states:

$$(\boldsymbol{\mu}_t^{\text{pri}}, \boldsymbol{\sigma}_t^{\text{pri}}) = f_{\text{trans}}(\mathbf{h}_{t-1}, \boldsymbol{\xi}_t), \qquad (\boldsymbol{\mu}_t^{\text{post}}, \boldsymbol{\sigma}_t^{\text{post}}) = f_{\text{post}}(\tilde{\mathbf{h}}_t, \mathbf{e}_t, \mathbf{c}_t). \tag{2}$$

Here $\tilde{\mathbf{h}}_t$ denotes the transition-produced time-t prior node state, from which the prior parameters $(\boldsymbol{\mu}_t^{\text{pri}}, \boldsymbol{\sigma}_t^{\text{pri}})$ are predicted. This prior state provides the node queries used to aggregate the current radar points into node evidence $\mathbf{e}_t$ and update gates $\mathbf{c}_t$. The posterior update then incorporates this current evidence through a confidence-aware update with inter-node coordination.

The temporal context $\boldsymbol{\xi}_t$ is computed only from the previous K posterior node states $\{\mathbf{h}_{t-K}, \ldots, \mathbf{h}_{t-1}\}$. For each past step $i \in \{t-K, \ldots, t-1\}$, we concatenate the node state with its first-order temporal difference, $[\mathbf{h}_i; \mathbf{h}_i - \mathbf{h}_{i-1}]$. At the lower window boundary $i = t-K$ the temporal difference is set to zero (equivalently, $\mathbf{h}_{t-K-1}$ is treated as $\mathbf{h}_{t-K}$). We then project each feature to a context space and aggregate the resulting sequence with a GRU, using the final hidden state as $\boldsymbol{\xi}_t$. Since $\boldsymbol{\xi}_t$ is computed before the current-frame radar point cloud is accessed, the predictive prior contains no information from the current observation. The posterior, in contrast, is explicitly conditioned on the current radar evidence through $\mathbf{e}_t$ and $\mathbf{c}_t$. Therefore, the corrective residual $\boldsymbol{\mu}_t^{\text{post}} - \boldsymbol{\mu}_t^{\text{pri}}$ reflects an observation-driven update to predicted motion rather than a direct re-encoding of the current point cloud. Appendices F.3, F.4, and F.5 provide the detailed temporal context encoder, context-conditioned transition module, and confidence-aware posterior update.

Latent-node grounding and readout. We ground the latent nodes in radar geometry by reconstructing the observed point cloud from the union of node decoders. Training uses a centered Chamfer observation loss [36] and a KL regularizer between the posterior and prior. The centering is applied only inside the reconstruction loss, while the encoder input remains in the original capture coordinates. These objectives encourage a stable predictive latent state rather than provide semantic supervision. After processing the full sequence, temporal attention pooling produces node summaries $\{\mathbf{n}^{(m)}\}_{m=1}^{M}$, and a permutation-invariant readout produces a global radar summary $\mathbf{g}_{\text{radar}}$.

3.2 Hidden Interaction as Observation-Supported Corrective Residuals

Our key premise is that decomposing radar motion into a predicted continuation and an observation-conditioned update yields a useful latent signal for interaction understanding. We treat the resulting correction as a predictive motion-update signal, rather than as a direct object/contact measurement or a magnitude-only interaction score. This signal summarizes how current radar observations revise each body-centric motion carrier relative to its predictive state, while remaining inherently ambiguous: large corrections may also arise from fast free motion, abrupt stopping, landing, balance recovery, viewpoint changes, or radar sparsity. Interaction semantics are therefore inferred only from temporally aggregated residual patterns and the captioning model, rather than from the residual magnitude itself. We therefore define the corrective residual of node m at time t as

$$\boldsymbol{\Delta}_t^{(m)} = \boldsymbol{\mu}_t^{\text{post},(m)} - \boldsymbol{\mu}_t^{\text{pri},(m)}. \tag{3}$$

The residual is not used as a standalone interaction score. Instead, residual histories are encoded over time and combined with node confidence and motion summaries before caption generation. Thus, the model uses temporal residual structure rather than an isolated residual magnitude.

3.3 Language-Aligned Semantic Bottleneck

Corrective residuals form a node-time sequence, with one residual vector $\boldsymbol{\Delta}_t^{(m)}$ for each latent node m and frame t. To make them useful for captioning, we encode each node's residual history with a shared deviation dynamics encoder. Given the residual sequence $\{\boldsymbol{\Delta}_t^{(m)}\}_{t=1}^{T}$, the encoder applies per-node one-dimensional temporal convolutions followed by masked temporal averaging:

$$\mathbf{u}^{(m)} = \text{MaskedMean}_t \left(f_{\text{conv}} \left(\{\boldsymbol{\Delta}_t^{(m)}\}_{t=1}^{T} \right) \right), \tag{4}$$

where f_{conv} denotes a shared temporal convolutional encoder applied independently to each node's residual history. Sequences are zero-padded to a fixed length for batching, and frames with no valid radar points are also masked out; hence temporal averages are computed only over valid frames. In our implementation, f_{conv} is a lightweight two-layer 1D temporal convolutional encoder. The resulting $\mathbf{u}^{(m)}$ is a compact residual-history code for node m.

We also compute two node-level selection statistics:

$$\bar{c}^{(m)} = \text{MaskedMean}_t \; c_t^{(m)}, \qquad \bar{\delta}^{(m)} = \text{MaskedMean}_t \left\| \boldsymbol{\Delta}_t^{(m)} \right\|_2. \tag{5}$$

The confidence statistic indicates whether the residual history of node m is supported by reliable radar evidence, while the magnitude statistic provides a coarse measure of whether the node receives

a non-trivial corrective update. These statistics are not used as standalone interaction scores; instead, they are concatenated with the residual-dynamics code to drive node-level attention pooling.

The semantic bottleneck is obtained by selecting informative nodes, fusing the selected residual-dynamics representation with the global radar summary, and projecting the result into a normalized language-aligned embedding:

$$\begin{aligned} \alpha^{(m)} &= \mathrm{softmax}_m \, \phi\Big([\mathbf{u}^{(m)}; \bar{c}^{(m)}; \bar{\delta}^{(m)}]\Big), \\ \mathbf{u}_{\mathrm{sel}} &= \sum_{m=1}^{M} \alpha^{(m)} \mathbf{u}^{(m)}, \\ \mathbf{s}_{\mathrm{shared}} &= f_{\mathrm{fuse}}\left([\mathbf{u}_{\mathrm{sel}}; \mathbf{g}_{\mathrm{radar}}]\right), \\ \mathbf{z}_{\mathrm{sem}} &= \ell_2\text{-normalize}\left(f_{\mathrm{sem}}(\mathbf{s}_{\mathrm{shared}})\right). \end{aligned} \tag{6}$$

Here ϕ is a small two-layer MLP that maps the per-node feature $[\mathbf{u}^{(m)}; \bar{c}^{(m)}; \bar{\delta}^{(m)}]$ to a scalar attention logit, f_{fuse} is a lightweight fusion MLP, and f_{sem} is a small two-layer MLP that projects into the language-aligned bottleneck dimension. The node attention weights are therefore conditioned on both learned residual dynamics and observation-support statistics. This allows the semantic bottleneck to select residual histories that are reliable and useful for captioning, without treating residual magnitude alone as interaction evidence.

3.4 Conditioning the Caption Decoder

We connect the radar representation to the language model through a structured prefix, following prefix-based conditioning of pretrained language models [37]. The prefix contains a task prompt, one semantic token derived from the language-aligned bottleneck, and node tokens projected from the node-level motion summaries:

$$\mathbf{\Pi} = [\mathbf{p}_{1:L_p}; \mathbf{q}; \mathbf{r}^{(1)}, \ldots, \mathbf{r}^{(M)}], \qquad \mathbf{q} = f_\theta(\mathbf{z}_{\mathrm{sem}}), \qquad \mathbf{r}^{(m)} = f_\psi(\mathbf{n}^{(m)}). \tag{7}$$

Here, f_θ and f_ψ are lightweight projection heads into the decoder embedding space.

This structured prefix provides two complementary conditioning pathways. The semantic token $\mathbf{q}$ carries the bottlenecked interaction-level summary, while the node tokens $\{\mathbf{r}^{(m)}\}$ retain node-level motion evidence from the observed radar sequence. This design helps separate interaction-level semantics from motion grounding: without the semantic token, the decoder has less access to a compact interaction-level summary and may over-rely on observed motion details; without node tokens, generated captions may be less grounded in the observed motion.

In practice, we use a frozen decoder-only language model with lightweight trainable adapters. The decoder is trained with the standard autoregressive captioning loss $\mathcal{L}_{\mathrm{cap}}$. To keep the semantic bottleneck active, we randomly drop node tokens during training, preventing the decoder from relying exclusively on fine-grained motion tokens while still allowing node-level details to refine the final description.

Training objective. The overall training objective combines latent-state grounding, variational regularization, cross-modal alignment, language-level semantic alignment, and caption generation:

$$\mathcal{L} = \lambda_{\mathrm{obs}}\mathcal{L}_{\mathrm{obs}} + \lambda_{\mathrm{KL}}\mathcal{L}_{\mathrm{KL}} + \lambda_{\mathrm{ctr}}\mathcal{L}_{\mathrm{ctr}} + \lambda_{\mathrm{sem}}\mathcal{L}_{\mathrm{sem}} + \lambda_{\mathrm{cap}}\mathcal{L}_{\mathrm{cap}}. \tag{8}$$

Here $\mathcal{L}_{\mathrm{obs}}$ grounds the node-structured latent state in radar geometry, $\mathcal{L}_{\mathrm{KL}}$ regularizes the observation-conditioned posterior toward the motion-continuation prior, $\mathcal{L}_{\mathrm{ctr}}$ aligns the global radar summary $\mathbf{g}_{\mathrm{radar}}$ with paired body-motion embeddings available from the synthesis pipeline; this paired motion branch is used only as auxiliary training supervision and is discarded at inference. Finally, $\mathcal{L}_{\mathrm{sem}}$ aligns the ℓ_2-normalized semantic bottleneck with ℓ_2-normalized caption-level language embeddings using a temperature-scaled contrastive loss, and $\mathcal{L}_{\mathrm{cap}}$ trains the decoder to generate the final caption. Together, these terms encourage the model to learn a predictive motion representation, summarize corrective residual information into a language-aligned semantic cue, and generate captions grounded in the observed radar sequence.

4 Experiments

4.1 Datasets and Evaluation Protocol

A fundamental data limitation in radar-based open-ended interaction understanding is the absence of paired radar point clouds and natural-language annotations. Existing radar datasets mainly provide closed-set action labels [1, 19, 3, 20], while language-paired motion datasets are not directly suitable for radar training. To address this limitation, we construct a paired synthetic radar–text corpus, inspired by RadarLLM [7] and simulation-based mmWave sensing data synthesis [38], and tailor it to our hidden-object interaction setting. We provide additional dataset-generation details in Appendix G.

We first create controlled motion specifications for object-absent motions and object-involving interactions, and convert them into ground-truth captions using a rule-based LLM prompting protocol. Each specification is then converted into 3D human motion with MDM [39], a text-conditioned human motion diffusion model, to generate a plausible 3D body-motion sequence. Thus, object-involving prompts affect the generated body motion only through textual conditioning; no explicit object meshes, trajectories, poses, or contact states are used.

The generated human motion is rendered into sparse mmWave radar point clouds using the RF-Genesis physics-based mmWave ray-tracing simulator [38], with radar parameters chosen to approximate our physical TI IWR6843AOP setting (see Appendix G for details). The final synthetic corpus contains 10,000 radar–caption clips, split into train, validation, and synthetic test sets an 80/10/10 ratio.

To evaluate synthetic-to-real transfer, we construct a real-world benchmark of 420 radar clips from two complementary sources. We first collect 180 clips using the same TI IWR6843AOP radar configuration as the synthetic renderer, covering two subjects, 15 object-involving motions, and 15 object-absent motions, each repeated three times. The object-involving clips contain physical objects in the scene, so the radar may include weak or intermittent object returns, but we do not provide object boxes, trajectories, contact labels, or object-level supervision.

We further include 240 object-absent real radar point-cloud clips from the mRI dataset [15] to evaluate transfer on real human actions without physical object interactions. Since mRI provides only action labels, we generate one natural-language caption per clip using a rule-based LLM prompting protocol conditioned on the action label and the ground-truth keypoint sequence. The keypoints are used only for offline caption annotation and are not provided to the model. In total, the real benchmark contains 420 clips and is used only for evaluation, with no real clip used for training, validation, hyperparameter tuning, or model selection.

4.2 Baselines and protocol

As noted by RadarLLM [7], directly comparable end-to-end radar-to-text baselines are scarce. We adopt a standardized two-stage evaluation protocol for non-radar captioning models. Specifically, we compare HIRaD against three families of methods: radar-driven (RD), skeleton-driven (SD), and video-driven (VD) baselines. Radar-driven (RD) baselines operate directly on radar point clouds. In contrast, SD and VD baselines do not natively consume radar signals, and therefore require an intermediate human-motion representation, whose quality is reported in Appendix B. For these baselines, we train a radar-to-motion reconstruction model using the motion annotations in the MDM training split as supervision. At evaluation time, no ground-truth motion annotations are used; the trained reconstruction model predicts skeletons or meshes from the radar sequence.

For SD baselines, the reconstructed skeleton sequence is fed to the motion tokenizers of MotionGPT [33] and AvatarGPT [40]. For VD baselines, we render the reconstructed motion as a video and apply video-capable vision-language models, including VideoLLaMA3 [41], Video-LLaVA [42], Qwen2.5-VL [43], and InternVL-2.5 [44]. For a controlled comparison, all captioning baselines are fine-tuned on the same training split and caption annotations as HIRaD, each using its native input modality. RD, SD, and HIRaD share the same frozen Qwen3-8B decoder backbone, QLoRA configuration, and validation-based model selection, so the differences among them reflect the conditioning representation rather than the language model. VD baselines are LoRA-fine-tuned under identical settings. Details are given in Appendix B. This protocol provides a consistent two-stage comparison for models that cannot directly process radar point clouds, while avoiding the use of ground-truth motion annotations at evaluation time.

We evaluate caption quality using four standard captioning metrics: CIDEr [45], METEOR [46], ROUGE-1 [47], and BLEU-1 [48]. To better assess interaction-relevant semantics, we additionally report SPICE [49], its object sub-score, and object-presence balanced accuracy (BAcc). These metrics complement surface-level word-overlap scores by evaluating semantic structure and whether the generated caption correctly preserves the presence or absence of an object.

5 Results

Main Captioning Results. We evaluate end-to-end radar captioning on synthetic and real radar benchmarks. All captioning results are averaged over three random seeds. As shown in Table 1, our method achieves the best overall performance. On synthetic data, it obtains the highest CIDEr, METEOR, ROUGE-1, and BLEU-1 scores, improving CIDEr from 4.3 to 7.2 over the radar-derived baseline RadarLLM. On real data, our method ranks first across all metrics, improving CIDEr from 2.2 to 4.8. These consistent gains indicate that observation-supported corrections improve both synthetic captioning performance and real-world transfer for radar-based interaction understanding.

Table 1: **Main comparison on synthetic captioning and real data captioning.** We report standard captioning metrics on both synthetic and real benchmarks. Our method achieves the best overall performance across both settings.

	Method	Synthetic Captioning				Real Captioning			
		CIDEr ↑	METEOR ↑	R-1 ↑	B-1 ↑	CIDEr ↑	METEOR ↑	R-1 ↑	B-1 ↑
SD†	MotionGPT	3.8±0.5	16.1±0.1	42.8±0.7	41.3±0.4	1.2±0.5	12.7±0.2	28.3±0.5	27.7±0.4
	AvatarGPT	3.9±0.5	16.6±0.1	43.1±0.4	42.5±0.5	1.1±0.2	13.4±0.1	29.2±0.5	31.7±0.8
VD‡	Video-LLaMA3	3.3±0.8	15.6±0.1	40.8±0.7	38.8±0.8	0.9±0.4	11.6±0.2	37.4±0.3	32.7±0.6
	Video-LLaVA	3.1±0.7	15.4±0.1	41.1±0.6	36.6±0.7	0.7±0.5	11.2±0.3	30.2±0.5	30.6±0.5
	Qwen2.5-VL	2.1±0.4	14.6±0.1	37.7±0.4	35.4±0.4	1.3±0.6	13.4±0.4	33.7±0.3	31.5±0.5
	InternVL-2.5	2.2±0.7	15.5±0.2	40.8±0.8	36.6±0.7	0.3±0.2	11.0±0.1	24.1±0.4	19.6±0.5
RD*	RadarLLM	4.3±0.5	16.4±0.2	42.7±0.3	42.4±0.5	2.2±0.5	13.8±0.2	35.4±0.4	33.5±0.5
	Ours	**7.2**±0.4	**18.5**±0.1	**43.8**±0.3	**45.5**±0.8	**4.8**±0.4	**16.5**±0.1	**41.1**±0.6	**35.9**±0.4

† Skeleton-derived. ‡ Video-derived. * Radar-derived.

Table 2: **Interaction benchmark on synthetic and real data.** We evaluate whether generated captions preserve interaction-level semantics using SPICE, its object sub-score, and object-presence balanced accuracy. Our method performs best overall, especially on semantic matching and object-presence prediction.

	Method	Synthetic interaction			Real interaction		
		SPICE ↑	Obj ↑	BAcc ↑	SPICE ↑	Obj ↑	BAcc ↑
SD†	MotionGPT	24.0±0.9	16.8±0.8	58.7±0.2	18.2±0.5	16.2±0.3	53.1±0.5
	AvatarGPT	30.5±0.7	28.3±0.9	72.0±0.3	22.3±0.7	20.8±0.3	63.4±0.5
VD‡	Video-LLaMA3	18.2±0.8	16.0±0.9	56.6±0.5	19.4±0.4	17.1±0.5	49.1±0.4
	Video-LLaVA	17.1±0.7	14.6±0.4	62.2±0.8	14.1±0.5	13.1±0.5	46.1±0.3
	Qwen2.5-VL	17.7±0.8	16.6±0.7	53.1±0.3	13.8±0.5	12.7±0.3	43.7±0.5
	InternVL-2.5	12.6±0.5	12.6±0.5	54.3±0.7	12.9±0.5	14.6±0.3	42.4±0.4
RD*	RadarLLM	33.5±0.6	31.2±0.7	62.5±0.3	20.3±0.4	19.1±0.4	58.9±0.8
	Ours	**41.7**±0.7	**40.4**±0.9	**72.9**±0.4	**29.9**±0.6	**27.6**±0.8	**65.3**±0.2

† Skeleton-derived. ‡ Video-derived. *Radar-derived.

Interaction Benchmark. We further evaluate whether generated captions preserve interaction-level semantics. As shown in Table 2, our method consistently improves over radar-, skeleton-, and video-derived baselines. On synthetic data, it improves SPICE from 33.5 to 41.7 and Obj from 31.2 to 40.4 over RadarLLM. On real data, it also obtains the highest SPICE, Obj, and BAcc scores, improving SPICE from 20.3 to 29.9, Obj from 19.1 to 27.6, and BAcc from 58.9 to 65.3. These

results indicate that our approach better recovers object-related interaction semantics, including under synthetic-to-real transfer.

Ablations. Table 3 summarizes the key ablation on the synthetic test split. Removing the radar prefix causes the largest drop, while latent tokens, motion supervision, and residual trajectories progressively improve CIDEr, SPICE, and Obj. The full model performs best, confirming that motion supervision and residual trajectories complement the latent encoder. Replacing the residual input to the semantic pathway with the non-subtracted posterior state, with everything else fixed, reduces CIDEr from 7.2 to 6.1, on par with the prior-only variant (4) (Appendix A). Zeroing the semantic token at inference reduces CIDEr to 4.7 both with and without residuals, so residual information reaches the decoder mainly through the semantic pathway (Appendix C).

Table 3: **Ablation on the synthetic test split.** Δ_{avg} is computed over CIDEr, SPICE, and Obj relative to the full model. Variant (2) corresponds to replacing our latent graph with the VQ-VAE-style codebook encoder used by RadarLLM [7].

#	Variant	Prefix	Codebook	Latent	Residual	Motion	CIDEr	SPICE	Obj	Δ_{avg}
(1)	No Radar Prefix	✗	✗	✗	✗	✗	0.8	13.3	11.5	-76.5%
(2)	Codebook Encoder	✓	✓	✗	✗	✓	4.3	33.5	31.2	-27.4%
(3)	w/o Motion Supervision	✓	✗	✓	✓	✗	4.9	37.9	36.2	-17.0%
(4)	w/o Residual Trajectory	✓	✗	✓	✗	✓	6.0	38.5	37.4	-10.6%
(5)	Full (Ours)	✓	✗	✓	✓	✓	**7.2**	**41.7**	**40.4**	–

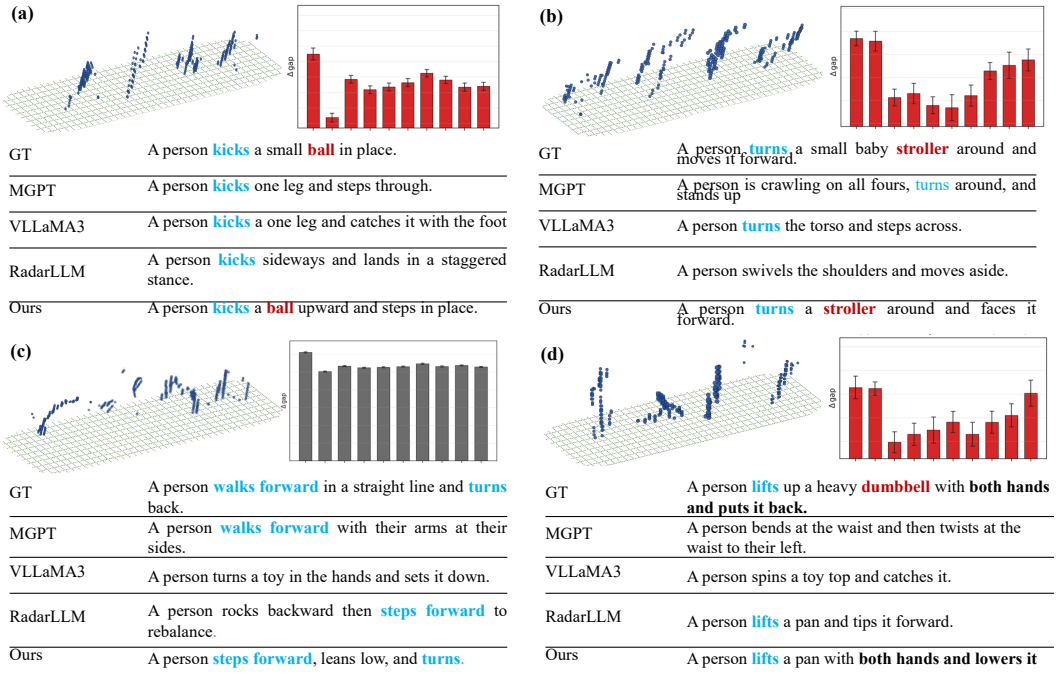


Figure 3: **Qualitative examples.** (a) and (b) show object-involving actions and their corresponding residual trajectories, and (c) shows a free-motion example for comparison. Compared with free motion, object-involving motion exhibits clearly different residual patterns, suggesting that the residual signal captures interaction-related motion cues beyond raw motion alone. (d) shows a similar tendency, although fine-grained object identity is not always correctly recognized.

Qualitative Analysis. Figure 3 shows representative qualitative examples. Our method often recovers object-involving interactions from body-motion dynamics even when object evidence in the radar point cloud is weak, intermittent, or absent. The examples suggest that the model does not rely on the hidden object's appearance, but instead learns how human motion is organized and

stabilized under object- or surface-related constraints. However, fine-grained object categories remain challenging, especially when different objects induce similar motions or produce sparse radar returns. The model is therefore better at capturing coarse interaction structure than precise object identity.

6 Conclusion

We presented HIRaD, a radar semantic sensing framework for hidden interaction understanding under weak object observability. Instead of directly detecting objects from sparse radar returns, our proposed method models how current observations correct a predicted body-motion prior and uses the resulting residual trajectories as interaction-relevant cues. By combining a node-structured predictive latent state, a language-aligned semantic bottleneck, and structured prefix conditioning of a frozen language model, HIRaD improves both synthetic captioning and real-world transfer. Experiments and ablations show that residual trajectories provide complementary information beyond latent motion features, while qualitative results indicate that they remain informative even when fine-grained object recognition is imperfect. These results suggest that predictive correction is a promising direction for radar-based open-ended interaction understanding.

7 Limitations

HIRaD shows that residual trajectories contain useful interaction cues, but these cues should not be interpreted as exact temporal contact or interaction boundaries. A large residual may also arise from abrupt free motion, balance recovery, or sparse radar observations (Appendix D), so it does not necessarily reflect object-induced constraints. We provide additional analysis in Appendix A. Our physical-object validation is limited in scale. Generalization to unseen objects and interactions, compositional settings, and paired counterfactual recordings of the same motion with and without the object remains unverified. In addition, our current setting assumes a single person in the scene, leaving multi-person interaction and occlusion handling for future work. Finally, the model is designed for relatively short clips. Extending it to longer sequences will require more scalable temporal modeling and more robust filtering of irrelevant motion changes. The prior/posterior formulation itself is modality-agnostic. Camera or LiDAR evidence could enter through modality-specific encoders with confidence-weighted fusion before the posterior update, which we leave to future work.

Acknowledgments and Disclosure of Funding

This work was supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP) grants funded by the Korea government (MSIT): RS-2024-00357879 (AI-based Biosignal Fusion and Generation Technology for Intelligent Personalized Chronic Disease Management), RS-2025-02219317 (AI Star Fellowship, Kookmin University), and IITP-2024-RS-2024-00417958 (Global Research Support Program in the Digital Field); and by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (RS-2026-25522067). This work was also supported by the AI Seoul Tech Research Support Program of the Seoul Future Foundation.

References

[1] Akash Deep Singh, Sandeep Singh Sandha, Luis Garcia, and Mani Srivastava. RadHAR: Human activity recognition from point clouds generated through a millimeter-wave radar. In *Proceedings of the 3rd ACM Workshop on Millimeter-wave Networks and Sensing Systems*, pages 51–56, 2019.

[2] Sujeet Milind Patole, Murat Torlak, Dan Wang, and Murtaza Ali. Automotive Radars: A review of signal processing techniques. *IEEE Signal Processing Magazine*, 34(2):22–35, 2017.

[3] Han Cui, Shu Zhong, Jiacheng Wu, Zichao Shen, Naim Dahnoun, and Yiren Zhao. MiliPoint: A point cloud dataset for mmwave radar. *Advances in Neural Information Processing Systems*, 36:62713–62726, 2023.

[4] Akash Deep Singh, Yunhao Ba, Ankur Sarker, Howard Zhang, Achuta Kadambi, Stefano Soatto, Mani Srivastava, and Alex Wong. Depth estimation from camera image and mmwave radar point cloud. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9275–9285, 2023.

[5] Kun Qian, Zhaoyuan He, and Xinyu Zhang. 3D point cloud generation with millimeter-wave radar. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 4(4):1–23, 2020.

[6] Yi Zhou, Lulu Liu, Haocheng Zhao, Miguel López-Benítez, Limin Yu, and Yutao Yue. Towards deep radar perception for autonomous driving: Datasets, methods, and challenges. *Sensors*, 22(11):4208, 2022.

[7] Zengyuan Lai, Jiarui Yang, Songpengcheng Xia, Lizhou Lin, Lan Sun, Renwen Wang, Jianran Liu, Qi Wu, and Ling Pei. RadarLLM: Empowering large language models to understand human motion from millimeter-wave point cloud sequence. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 5791–5799, 2026.

[8] Youngwook Kim, Ibrahim Alnujaim, and Daegun Oh. Human activity classification based on point clouds measured by millimeter wave mimo radar with deep recurrent neural networks. *IEEE Sensors Journal*, 21 (12):13522–13529, 2021.

[9] Zhanjun Hao, Jiaxing Xiao, Yuejiao Wang, Guowei Wang, and Fenfang Li. mm-ARnet: Exploring millimeter wave radar point clouds for human action recognition. *IEEE Transactions on Mobile Computing*, 2025.

[10] James J Gibson. *The ecological approach to visual perception: classic edition*. Psychology Press, 2014.

[11] Rajesh PN Rao and Dana H Ballard. Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1):79–87, 1999.

[12] Arindam Sengupta, Feng Jin, Renyuan Zhang, and Siyang Cao. mm-Pose: Real-time human skeletal posture estimation using mmwave radars and cnns. *IEEE Sensors Journal*, 20(17):10032–10044, 2020.

[13] Hongfei Xue, Yan Ju, Chenglin Miao, Yijiang Wang, Shiyang Wang, Aidong Zhang, and Lu Su. mmMesh: Towards 3d real-time dynamic human mesh construction using millimeter-wave. In *Proceedings of the 19th Annual International Conference on Mobile Systems, Applications, and Services*, pages 269–282, 2021.

[14] Hehe Fan, Yi Yang, and Mohan Kankanhalli. Point 4d transformer networks for spatio-temporal modeling in point cloud videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14204–14213, 2021.

[15] Sizhe An, Yin Li, and Umit Ogras. mRI: Multi-modal 3d human pose estimation dataset using mmwave, rgb-d, and inertial sensors. *Advances in Neural Information Processing Systems*, 35:27414–27426, 2022.

[16] Xiaopeng Zhao, Zhenlin An, Qingrui Pan, and Lei Yang. NeRF2: Neural radio-frequency radiance fields. In *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking*, pages 1–15, 2023.

[17] David Borts, Erich Liang, Tim Broedermann, Andrea Ramazzina, Stefanie Walz, Edoardo Palladin, Jipeng Sun, David Brueggemann, Christos Sakaridis, Luc Van Gool, et al. Radar fields: Frequency-space neural scene representations for fmcw radar. In *ACM SIGGRAPH 2024 Conference Papers*, pages 1–10, 2024.

[18] Jiachen Lu, Hailan Shanbhag, and Haitham Al Hassanieh. GeRaF: Neural geometry reconstruction from radio frequency signals. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.

[19] Sameera Palipana, Dariush Salami, Luis A Leiva, and Stephan Sigg. Pantomime: Mid-air gesture recognition with sparse millimeter-wave radar point clouds. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 5(1):1–27, 2021.

[20] Jianfei Yang, He Huang, Yunjiao Zhou, Xinyan Chen, Yuecong Xu, Shenghai Yuan, Han Zou, Chris Xiaoxuan Lu, and Lihua Xie. MM-Fi: Multi-modal non-intrusive 4d human dataset for versatile wireless sensing. *Advances in Neural Information Processing Systems*, 36:18756–18768, 2023.

[21] Yu-Wei Chao, Yunfan Liu, Xieyang Liu, Huayi Zeng, and Jia Deng. Learning to detect human-object interactions. In *2018 IEEE Winter Conference on Applications of Computer Vision*, pages 381–389. IEEE, 2018.

[22] Georgia Gkioxari, Ross Girshick, Piotr Dollár, and Kaiming He. Detecting and recognizing human-object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8359–8367, 2018.

[23] Zhi Hou, Baosheng Yu, Yu Qiao, Xiaojiang Peng, and Dacheng Tao. Affordance transfer learning for human-object interaction detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 495–504, 2021.

[24] Lin Wu, Zhixiang Chen, and Jianglin Lan. HOI-Dyn: Learning interaction dynamics for human-object motion diffusion. *arXiv preprint arXiv:2507.01737*, 2025.

[25] Zerui Chen, Rolandos Alexandros Potamias, Shizhe Chen, Jiankang Deng, Cordelia Schmid, and Stefanos Zafeiriou. HO-Flow: Generalizable hand-object interaction generation with latent flow matching. *arXiv preprint arXiv:2604.10836*, 2026.

[26] Francesco Locatello, Dirk Weissenborn, Thomas Unterthiner, Aravindh Mahendran, Georg Heigold, Jakob Uszkoreit, Alexey Dosovitskiy, and Thomas Kipf. Object-centric learning with slot attention. *Advances in Neural Information Processing Systems*, 33:11525–11538, 2020.

[27] Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning*, pages 2555–2565. PMLR, 2019.

[28] Malte Mosbach, Jan Niklas Ewertz, Angel Villar-Corrales, and Sven Behnke. SOLD: Slot object-centric latent dynamics models for relational manipulation learning from pixels. *arXiv preprint arXiv:2410.08822*, 2024.

[29] Seokha Moon, Hongbeen Park, Jaekoo Lee, and Jinkyu Kim. Learning temporal cues by predicting objects move for multi-camera 3d object detection. In *2024 IEEE International Conference on Robotics and Automation*, pages 6607–6613. IEEE, 2024.

[30] Chuan Guo, Inwoo Hwang, Jian Wang, and Bing Zhou. SnapMoGen: Human motion generation from expressive texts. In *Advances in Neural Information Processing Systems*, volume 38, pages 99939–99955, 2025.

[31] Zhiwei Zheng, Shibo Jin, Lingjie Liu, and Mingmin Zhao. Next-scale autoregressive models for text-to-motion generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16376–16386, June 2026.

[32] KunHo Heo, SuYeon Kim, Yonghyun Gwon, Youngbin Kim, and MyeongAh Cho. ParTY: Part-guidance for expressive text-to-motion synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23549–23558, 2026.

[33] Biao Jiang, Xin Chen, Wen Liu, Jingyi Yu, Gang Yu, and Tao Chen. MotionGPT: Human motion as a foreign language. *Advances in Neural Information Processing Systems*, 36:20067–20079, 2023.

[34] Chuan Guo, Xinxin Zuo, Sen Wang, and Li Cheng. TM2T: Stochastic and tokenized modeling for the reciprocal generation of 3d human motions and texts. In *European Conference on Computer Vision*, pages 580–597. Springer, 2022.

[35] Ling-Hao Chen, Shunlin Lu, Ailing Zeng, Hao Zhang, Benyou Wang, Ruimao Zhang, and Lei Zhang. MotionLLM: Understanding human behaviors from human motions and videos. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.

[36] Haoqiang Fan, Hao Su, and Leonidas J Guibas. A point set generation network for 3d object reconstruction from a single image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 605–613, 2017.

[37] Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 4582–4597, 2021.

[38] Xingyu Chen and Xinyu Zhang. RF-genesis: Zero-shot generalization of mmwave sensing through simulation-based data synthesis and generative diffusion models. In *Proceedings of the 21st ACM Conference on Embedded Networked Sensor Systems*, pages 28–42, 2023.

[39] Guy Tevet, Sigal Raab, Brian Gordon, Yoni Shafir, Daniel Cohen-or, and Amit Haim Bermano. Human motion diffusion model. In *The Eleventh International Conference on Learning Representations*, 2023.

[40] Zixiang Zhou, Yu Wan, and Baoyuan Wang. AvatarGPT: All-in-one framework for motion understanding planning generation and beyond. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1357–1366, 2024.

[41] Boqiang Zhang, Kehan Li, Zesen Cheng, Zhiqiang Hu, Yuqian Yuan, Guanzheng Chen, Sicong Leng, Yuming Jiang, Hang Zhang, Xin Li, et al. VideoLLaMA 3: Frontier multimodal foundation models for image and video understanding. *arXiv preprint arXiv:2501.13106*, 2025.

[42] Bin Lin, Yang Ye, Bin Zhu, Jiaxi Cui, Munan Ning, Peng Jin, and Li Yuan. Video-LLaVA: Learning united visual representation by alignment before projection. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 5971–5984, 2024.

[43] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025. doi: 10.48550/arXiv.2502.13923.

[44] Zhe Chen, Weiyun Wang, Yue Cao, Yangzhou Liu, Zhangwei Gao, Erfei Cui, Jinguo Zhu, Shenglong Ye, Hao Tian, Zhaoyang Liu, et al. Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling. *arXiv preprint arXiv:2412.05271*, 2024.

[45] Ramakrishna Vedantam, C Lawrence Zitnick, and Devi Parikh. CIDER: Consensus-based image description evaluation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4566–4575, 2015.

[46] Satanjeev Banerjee and Alon Lavie. METEOR: An automatic metric for mt evaluation with improved correlation with human judgments. In *Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization*, pages 65–72, 2005.

[47] Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, 2004.

[48] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. BLEU: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, page 311–318, 2002.

[49] Peter Anderson, Basura Fernando, Mark Johnson, and Stephen Gould. SPICE: Semantic propositional image caption evaluation. In *European Conference on Computer Vision*, page 382–398. Springer, 2016.

[50] Bharat Lal Bhatnagar, Xianghui Xie, Ilya A Petrov, Cristian Sminchisescu, Christian Theobalt, and Gerard Pons-Moll. Behave: Dataset and method for tracking human object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15935–15946, 2022.

[51] Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.

[52] Daehee Kim, Youngjun Yoo, Seunghyun Park, Jinkyu Kim, and Jaekoo Lee. SelfReg: Self-supervised contrastive regularization for domain generalization. In *2021 IEEE/CVF International Conference on Computer Vision*, pages 9619–9628. IEEE, 2021.

A Analysis of Corrective Residuals as Interaction Cues

We further analyze whether the corrective residuals used by our model contain interaction-relevant structure, rather than only reflecting generic motion variation. Our goal is not to treat the residual as a direct measurement of object contact or object presence. Instead, we ask whether the prior–posterior correction induced by radar observations exhibits temporal and directional patterns that are informative for distinguishing object-involving motions from object-absent motions with similar coarse body dynamics.

Temporal gap dynamics. For each frame, we define the node-averaged prior–posterior gap as

$$\mathrm{Gap}_t = \frac{1}{M} \sum_{m=1}^{M} \left\| \boldsymbol{\mu}_t^{\mathrm{post},(m)} - \boldsymbol{\mu}_t^{\mathrm{pri},(m)} \right\|_2, \tag{9}$$

where $\boldsymbol{\mu}_t^{\mathrm{pri},(m)}$ and $\boldsymbol{\mu}_t^{\mathrm{post},(m)}$ denote the prior and posterior means of node m, respectively. This quantity measures how much the predicted latent state is corrected after observing the current radar frame. Since a large gap can also arise from abrupt free motion, landing, balance recovery, or noisy radar returns, we do not interpret a single-frame gap as evidence of an interaction. We instead compare temporal gap patterns across matched groups.

We analyze 720 clips from the synthetic validation split, consisting of 420 object-involving clips and 300 object-absent hard negatives. The object-involving clips contain captions with both an interaction verb and a concrete object word. The hard negatives contain no concrete object word, but are selected from similar coarse motion categories such as bending, squatting, leaning, arm swinging, kicking, and hopping. This construction reduces the chance that the analysis merely separates object interactions from unrelated body motions.

Because exact contact times are not annotated, we align clips by normalized time and summarize each clip using the change in node-averaged gap:

$$\Delta_{\mathrm{gap}} = \mathrm{Mean}_{t \in \mathcal{W}_{\mathrm{late}}} \mathrm{Gap}_t - \mathrm{Mean}_{t \in \mathcal{W}_{\mathrm{early}}} \mathrm{Gap}_t. \tag{10}$$

Here, $\mathcal{W}_{\mathrm{early}}$ and $\mathcal{W}_{\mathrm{late}}$ are early and later windows on the normalized timeline. A negative value indicates that the prior–posterior gap decreases over the clip.

Figure 4 shows four matched example pairs. Each pair compares an object-involving motion with an object-absent hard negative that has similar coarse body dynamics: picking up an object versus free bending, lifting a cup versus raising both arms, opening a drawer versus a free cross-body reach, and swinging a racket versus a free torso swing. For each clip, we plot the mean node-averaged gap in four equal-length temporal bins. The object-involving examples tend to show a larger early correction followed by lower later bins, whereas the matched object-absent examples are flatter or slightly increasing. This suggests that the informative signal is not simply a larger residual magnitude, but a temporal correction pattern.

Table 4 reports the aggregate statistics over the same 720 clips. Object-involving motions have a more negative mean gap change (-0.213) than object-absent hard negatives (-0.014). In addition, 69.5% of object-involving clips have negative Δ_{gap}, compared with 54.0% of hard negatives. These results indicate that corrective residuals capture temporal structure correlated with object-involving interactions, although the overlap between groups also shows that the gap alone is not a standalone interaction detector.

Table 4: **Gap-change statistics for object-involving motions and hard negatives.** Negative Δ_{gap} indicates that the prior–posterior gap decreases from the early window to the later window. Object-involving motions show a more negative mean gap change and a higher fraction of negative gap changes than object-absent hard negatives with similar coarse motion categories.

Group	Mean $\Delta_{\mathrm{gap}} \downarrow$	Std.	% Negative $\uparrow$
Object-absent hard negatives	-0.014	0.462	54.0
Object-involving motions	$\mathbf{-0.213}$	0.446	$\mathbf{69.5}$

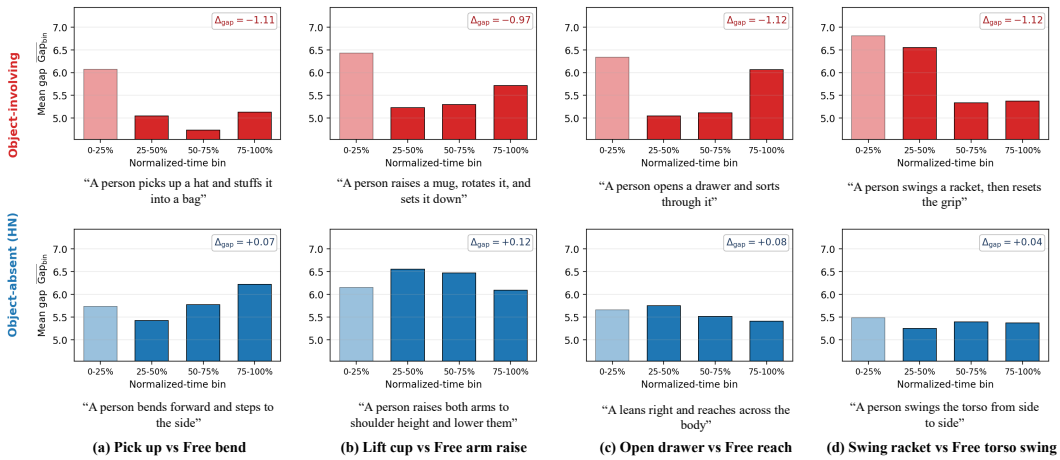


Figure 4: **Prior–posterior gap dynamics for matched object-involving and object-absent examples.** Each object-involving clip is paired with a hard negative that has similar coarse body motion: pick up vs. free bend, lift cup vs. free arm raise, open drawer vs. free cross-body reach, and swing racket vs. free torso swing. Object-involving examples show a larger early correction followed by lower later bins, while hard negatives are flatter or slightly increasing. This supports the interpretation of the residual as a temporal interaction cue, rather than merely a measure of overall motion intensity.

Directional residual structure. The preceding analysis only uses the residual norm. We therefore also examine whether the residual vectors contain directional structure beyond magnitude. For each of the same 720 clips, we construct a residual signature by averaging the vector residuals over an interaction window and concatenating them across nodes:

$$\mathbf{s} = \left[\overline{\boldsymbol{\Delta}}^{(1)}; \ldots; \overline{\boldsymbol{\Delta}}^{(M)}\right] \in \mathbb{R}^{M \cdot D}, \quad \overline{\boldsymbol{\Delta}}^{(m)} = \underset{t \in \mathcal{W}_{\text{int}}}{\text{Mean}} \boldsymbol{\Delta}_t^{(m)}. \tag{11}$$

We project these signatures into two dimensions using cosine-metric t-SNE. Because cosine geometry reduces sensitivity to clip-level scale, separation in this embedding is less directly attributable to residual magnitude alone.

Figure 5 shows the resulting embedding. Object-involving and hard-negative clips occupy partially distinct regions, with silhouette score $+0.18$ and k-NN-10 purity 0.77, compared with 0.58 chance. This suggests that the corrective residuals contain directional structure correlated with object-involving interactions. However, the same representation does not meaningfully separate heavy and light object interactions within the object-involving subset: the silhouette score is only $+0.01$, and the k-NN-10 purity is 0.49, close to the 0.48 chance level. We include this negative result to clarify the scope of the residual cue: it is useful as an interaction-relevant motion-update signal, but it is not sufficient by itself to recover fine-grained object properties such as mass class.

Task-level and real-data controls. On the held-out synthetic test split, we use the same matched construction as above (205 object-involving / 221 hard-negative clips matched by coarse motion category). A class-balanced logistic probe on frozen per-clip features (chance 50.0 BAcc) gives 86.9 BAcc / 93.9 AUC for residual trajectories, 86.2 / 93.3 for posterior states, and 85.2 / 92.6 for the motion prior. At the caption level, the full model names the correct object in 25.4% of object-involving clips, compared with 14.6% for the w/o-Residual ablation (Table 3, variant 4). We then repeat the analysis on the 180 directly collected real clips (two subjects, 15 object-involving and 15 object-absent motions, three repetitions each), with the encoder trained purely on synthetic data and kept frozen and the real recordings never passing through MDM, captions, or the language model. A linear probe for physical object presence, cross-validated with folds grouped by action type (representation-level BAcc, chance $= 50$), reaches 75.8 on the motion-continuation prior, 79.0 on posterior states, and 82.3 on residual trajectories. On a motion-matched subset (bend, sit, lift, turn, and step families with and without a physical object), the probe reaches 78.1. The separation

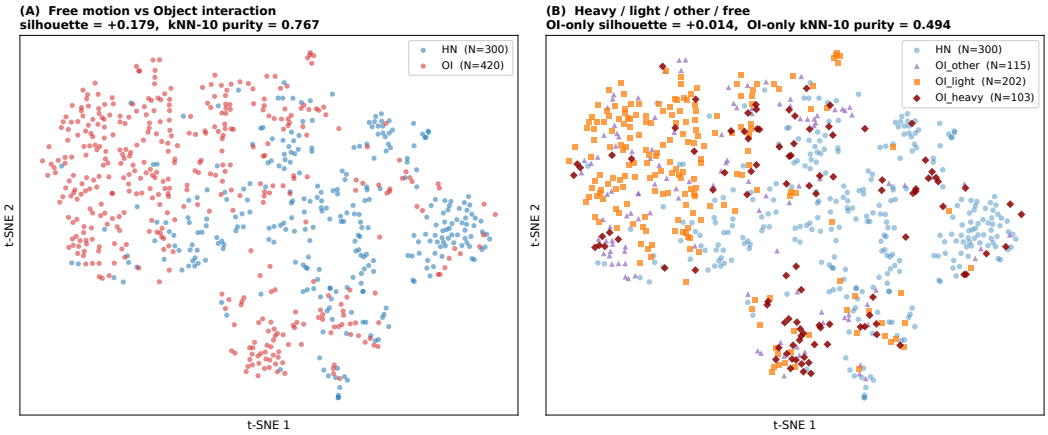


Figure 5: **Cosine t-SNE of time-averaged corrective residual signatures** ($N = 720$)**.** *(A)* Object-involving (OI) and hard-negative (HN) clips form partially separated regions, indicating directional residual structure beyond magnitude alone. *(B)* Heavy and light object interactions are not meaningfully separated. The residual signature therefore provides an interaction-relevant cue, but not a standalone descriptor of object properties.

therefore holds on recordings in which objects physically constrain the motion, independently of the synthetic motion generator.

Prediction-subtraction control. We also test whether the subtraction itself matters. We train a control variant in which the input to the semantic pathway is the non-subtracted observation-corrected state $\boldsymbol{\mu}_t^{\text{post}}$ instead of $\boldsymbol{\Delta}_t = \boldsymbol{\mu}_t^{\text{post}} - \boldsymbol{\mu}_t^{\text{pri}}$. Both have the same shape, so the architecture, parameter count, and capacity are identical. The variant is retrained from scratch under the same three-stage protocol, hyperparameters, data, and splits, and evaluated on the held-out synthetic test split. The prior-only variant ($\boldsymbol{\mu}_t^{\text{pri}}$ as input) corresponds to variant (4) of Table 3. The control reaches 6.07 CIDEr, against 7.2 for the full model and 6.02 for the prior-only variant, and the same ordering holds on BLEU-1, METEOR, and ROUGE-1. Teacher-forced validation losses are essentially identical across the three (1.289 / 1.280 / 1.277), so the gap emerges in free-running generation rather than in optimization. Encoding the complete corrected state therefore does not recover the benefit of explicitly representing the prior–posterior difference.

Overall, this analysis supports the use of corrective residuals as auxiliary interaction cues for radar captioning. The residual should not be interpreted as a direct sensor-level observation of an object, contact event, or object attribute. Rather, its temporal and directional structure provides complementary evidence that can be integrated with raw radar motion features by the captioning model.

B Baseline Comparability

Skeleton- and video-conditioned baselines. The Skeleton-Driven (SD) and Video-Driven (VD) baselines in the main comparison use an explicit reconstruction stage before captioning. Specifically, the input radar point cloud is first converted into a human pose sequence. For VD, the recovered pose is further rendered as a body mesh video. The resulting intermediate representation is then captioned by an off-the-shelf skeleton-conditioned or video-conditioned model. Therefore, the performance of SD and VD depends on both the radar-to-body reconstruction quality and the downstream captioning model.

To provide temporally stable skeleton and mesh inputs for the SD and VD baselines, we use mmMesh [13] radar-to-body reconstruction front-end. This front-end regresses per-frame SMPL parameters from the 4-channel point cloud and recovers 24 body joints through a shape-aware forward-kinematics head. The same reconstruction model provides joints for SD and meshes for VD, ensuring that the two baselines differ only in the downstream captioning modality. The front-end is

trained on the same training split with motion supervision taken only from the synthesis pipeline, and no ground-truth motion is used at evaluation. On top of it, SD baselines use the native motion tokenizers of MotionGPT and AvatarGPT with lightweight input projections, RD uses RadarLLM's motion-guided tokenizer with its published recipe, and VD baselines are LoRA-fine-tuned on rendered mesh video. All three families and HIRaD condition the same frozen Qwen3-8B decoder through QLoRA with identical adapter configuration and validation-based checkpoint selection, so the gaps among them isolate the conditioning representation rather than language-model capacity.

We further evaluate this choice against standard radar keypoint regressors. As a sanity check, we train the three official MiliPoint [3] keypoint backbones, PointNet, DGCNN, and PointTransformer, on the same training data. These models are used as per-frame xyz-only regressors, without an SMPL head or temporal smoothness term. Table 5 reports reconstruction quality on the held-out test split, evaluated in the body-centric frame on the first 22 SMPL body joints.

PointNet achieves the lowest per-frame MPJPE in Table 5. However, its trajectories exhibit substantially larger velocity and acceleration errors than our front-end. Our reconstruction front-end provides comparable per-frame pose accuracy while producing smoother motion trajectories. This trade-off is important for the SD and VD baselines because downstream captioning depends on temporal motion cues, not only on isolated per-frame pose accuracy. We therefore use our front-end as the reconstruction module for SD and VD. This choice does not make reconstruction-based baselines identical to radar-direct captioning, but it reduces the likelihood that their performance is limited by temporal jitter in the intermediate skeleton or mesh. Figure 6 shows representative reconstructions spanning the MPJPE distribution.

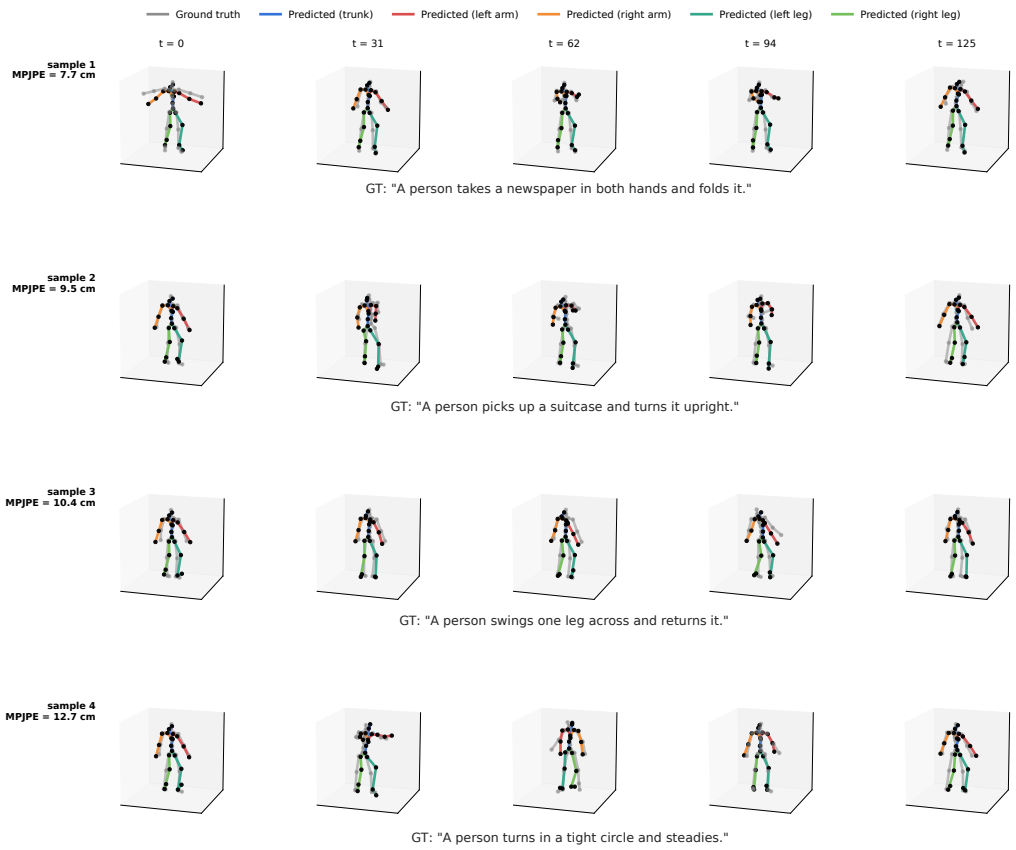


Figure 6: Representative SMPL reconstructions across the MPJPE distribution. Low-, medium-, and high-error examples show that the intermediate mesh stage preserves temporally coherent body motion, reducing the likelihood that captioning performance is limited by skeleton or mesh jitter.

Table 5: **Radar-to-skeleton reconstruction quality on the held-out test split.** Errors are evaluated in the body-centric frame using the first 22 SMPL body joints. Ours denotes our radar-to-body reconstruction front-end used to construct the SD and VD baselines. While PointNet obtains the lowest per-frame MPJPE, our front-end produces substantially smoother motion trajectories, as reflected by lower velocity and acceleration errors.

Method	MPJPE (cm)	Median	Vel. (cm/f)	Acc. (cm/f^2)
Ours (mmMesh)	11.74 ± 4.57	9.50	$\mathbf{1.36 \pm 0.81}$	$\mathbf{1.18 \pm 0.74}$
MiliPoint – PointNet	$\mathbf{10.35 \pm 6.11}$	**9.02**	5.12 ± 2.09	8.41 ± 3.47
MiliPoint – DGCNN	11.60 ± 6.77	10.10	4.33 ± 2.02	6.98 ± 3.15
MiliPoint – PointTransformer	11.69 ± 6.45	10.23	5.11 ± 2.28	8.36 ± 3.73

Effect of explicit object meshes in simulated radar. Our real-world dataset contains objects in the scene whereas our synthetic training data does not. To verify that this domain gap is negligible at the input level, we ran a controlled comparison on the BEHAVE human–object interaction dataset [50]. For 10 sequences spanning 10 object categories (*backpack*, *boxlarge*, *chairblack*, *chairwood*, *yogaball*, *suitcase*, *tablesquare*, *monitor*, *trashbin*, *yogamat*), we used the provided per-frame SMPL fits and 6-DoF object poses to construct two scenes per frame: one containing both the human mesh and the fitted object mesh, and one containing only the human mesh. Both scenes were rendered through the RF-Genesis pathtracer [38] with an identical sensor placement and processed through the IWR6843AOP radar signal generator with CA-CFAR detection, yielding 18 sparse mmWave point clouds per sequence and per condition.

Aggregated across the full sequence (Fig. 7), the two conditions are essentially indistinguishable. On average a frame produces 42.9 points with the object versus 40.3 without, and the mean per-frame point-count difference is only $|\Delta n| = 7.7$. The symmetric Chamfer distance between the two point clouds averages 7.0 cm — well within typical mmWave range/angle resolution. Even rigid objects with large radar cross sections (*yogaball*, *tablesquare*, chairs) leave no visually distinguishable signature in the post-CFAR point cloud. We attribute this to the inherent sparsity of mmWave sensing: a typical IWR6843AOP frame retains only a few dozen detections after CFAR, so the object's contribution is suppressed below the detection threshold or is occluded by the body's stronger return. This supports our use of object-free synthetic data as a practical training approximation for the sparse post-CFAR point-cloud regime, while real-object effects are still evaluated in the real benchmark.

C Semantic Bottleneck Analysis

All analyses use the held-out synthetic test split, with one seed per retrained variant. Retraining with $\lambda_{\text{sem}} = 0$ reduces CIDEr from 7.2 to 5.81 and correct-object naming on the matched object-involving subset from 25.4% to 15.1%, while the object-mention rate barely moves (59.5% to 56.6%). Retraining without the semantic token $\mathbf{q}$, with $\mathcal{L}_{\text{sem}}$ still active, reaches a comparable floor (CIDEr 5.39, correct-object 15.1%). The bottleneck therefore needs both the aligned representation and the channel that delivers it.

A linear probe on $\mathbf{z}_{\text{sem}}$ classifies 8 interaction sub-types at top-1 53.3% / top-2 80.0% / top-3 90.0% (chance 12.5%), matching or exceeding every 2560-d internal feature pool, and 80% of its errors stay within the correct object-involving/object-absent group. Nearest captions in $\mathbf{z}_{\text{sem}}$ space are semantically coherent ("A person bends at the waist and unbends." $\rightarrow$ "A person bows slightly and straightens."), whereas instance-level caption retrieval is near chance, so the bottleneck carries category-level interaction semantics. Zeroing $\mathbf{q}$ at inference reduces CIDEr from 7.2 to 4.71 and more than doubles object hallucination on object-absent clips (12.7% to 29.0%), and the same zeroing applied to the w/o-Residual variant (Table 3, variant 4) gives 4.74, so the two variants converge once the semantic token is removed. Zeroing the node tokens instead is catastrophic (0.94). Residual information thus reaches the decoder through the semantic pathway, while node tokens provide motion grounding.



Figure 7: Sparse mmWave point clouds simulated from BEHAVE [50] sequences with the human mesh alone (blue, bottom rows) versus human + fitted object mesh (red, top rows). Each panel overlays all 18 radar frames of one sequence on the SMPL silhouette (gray points) and the object's axis-aligned bounding box (gray box). Across all 10 object categories the two conditions are visually and quantitatively indistinguishable (mean $|\Delta n| = 7.7$ points/frame; mean symmetric Chamfer distance $= 7.0$ cm).

D Robustness Under Degraded Sensing

Table 6 stress-tests the trained model on the synthetic test split under point dropout, spatial jitter, Doppler noise, multipath ghost points, and lower-body occlusion. Under moderate degradation (20% dropout, 5 cm jitter, Doppler noise, ghost points, 50% occlusion) performance stays within 0.09 CIDEr, 0.2 SPICE, and 0.3 Obj of clean, and even at 80% dropout Obj (34.2) exceeds the strongest baseline's clean score (RadarLLM 31.2, Table 2). The gate measures routing reliability rather than point count, so it is preserved under dropout and decreases only under geometry-distorting jitter, while the residual norm rises with corruption severity, which is why residual magnitude is never used as an interaction score (Sec. 3.2).

Table 6: **Robustness to sensing degradation** (synthetic test split). Gate values are averaged over non-empty frames.

Condition	CIDEr	SPICE	Obj	gate $\bar{c}$	$\|\boldsymbol{\Delta}\|$
Clean	7.20	41.7	40.4	0.926	5.75
Dropout 20 / 50 / 80%	7.14 / 6.87 / 4.54	41.6 / 40.0 / 35.6	40.3 / 38.8 / 34.2	0.926 / 0.926 / 0.923	5.78 / 5.93 / 6.38
Jitter 5 / 15 cm	7.17 / 4.60	41.6 / 33.3	40.4 / 31.0	0.923 / 0.904	5.78 / 6.08
Doppler noise 0.2	7.14	41.5	40.3	0.927	5.76
Ghost points 20%	7.28	41.9	40.7	0.920	5.80
Occlusion 50%	7.11	41.5	40.1	0.929	5.78

E Additional Qualitative Results

Figure 8 presents additional qualitative examples from the real test set. Consistent with the main results, different methods can capture coarse human motions reasonably well. However, when an object is involved, our method produces captions that are more faithful to the interaction context and better preserve object-related semantics.

These examples suggest that the proposed latent and residual representations provide useful cues beyond coarse body motion alone, especially for object-involved interactions. Nevertheless, fine-grained action understanding remains challenging: the model may capture the overall interaction pattern, but still confuse subtle action details or precise object categories. This points to an important direction for future improvement.

F Method Detail

This section provides method details for components that are summarized only briefly in the main paper. We describe the multi-head point-to-node routing for node evidence, the routing-derived update gate used in the posterior update, and the temporal context construction that prevents current-frame leakage into the prior.

F.1 Point-to-Node Routing and Node Evidence

Let $\mathbf{p}_t^{(n)}$ be radar point n at frame t, and let $\mathbf{x}_t^{(n)} = f_{\text{obs}}(\mathbf{p}_t^{(n)})$ denote its encoded point feature produced by the shared masked point encoder. We use $\eta_t^{(n)} \in \{0, 1\}$ to denote the valid-point mask. Point features are routed to latent nodes using queries derived from the prior node states $\tilde{\mathbf{h}}_t^{(m)}$, which are predicted from past node states before the current radar frame is observed.

To preserve per-node identity in the query subspace, we add a learnable node-specific bias $\mathbf{b}^{(m)} \in \mathbb{R}^{d_{\text{node}}}$ before the query projection. For each attention head $h \in \{1, \ldots, H\}$, we compute node queries, point keys, and point values as

$$\mathbf{q}_{t,h}^{(m)} = W_{q,h}\left(\tilde{\mathbf{h}}_t^{(m)} + \mathbf{b}^{(m)}\right), \qquad \mathbf{k}_{t,h}^{(n)} = W_{k,h}\mathbf{x}_t^{(n)}, \qquad \mathbf{v}_{t,h}^{(n)} = W_{v,h}\mathbf{x}_t^{(n)}. \tag{12}$$

The pre-softmax routing logit between node m and point n is computed by scaled dot-product compatibility:

$$\ell_t^{(m,n,h)} = \frac{\left\langle \mathbf{q}_{t,h}^{(m)}, \mathbf{k}_{t,h}^{(n)} \right\rangle}{\sqrt{d_h}}. \tag{13}$$

Invalid or padded points are masked before normalization.

We compute two masked routing distributions. The first normalizes over nodes for each valid point, making nodes compete to explain the point:

$$a_{t,\text{node}}^{(m,n,h)} = \eta_t^{(n)} \frac{\exp(\ell_t^{(m,n,h)})}{\sum_{m'=1}^{M} \exp(\ell_t^{(m',n,h)}) + \epsilon}. \tag{14}$$

The second normalizes over valid points for each node, encouraging each node to attend to a coherent subset of the current observation:

$$a_{t,\text{point}}^{(m,n,h)} = \frac{\eta_t^{(n)} \exp(\ell_t^{(m,n,h)})}{\sum_{j=1}^{N} \eta_t^{(j)} \exp(\ell_t^{(m,j,h)}) + \epsilon}. \tag{15}$$

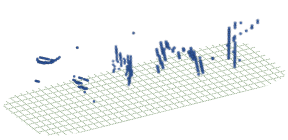
The final head-specific routing weight is the convex combination

$$a_t^{(m,n,h)} = (1 - \alpha)\, a_{t,\text{point}}^{(m,n,h)} + \alpha\, a_{t,\text{node}}^{(m,n,h)}, \tag{16}$$

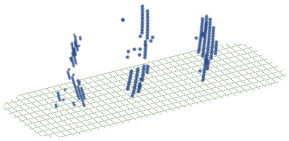
where α corresponds to the competition weight used in the implementation.

Using these head-specific routing weights, node evidence is obtained by multi-head value aggregation:

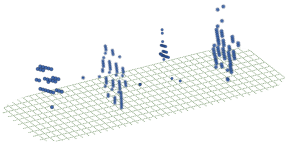
$$\mathbf{o}_{t,h}^{(m)} = \sum_{n=1}^{N} a_t^{(m,n,h)} \mathbf{v}_{t,h}^{(n)}, \qquad \mathbf{e}_t^{(m)} = W_o\left[\mathbf{o}_{t,1}^{(m)}; \ldots; \mathbf{o}_{t,H}^{(m)}\right]. \tag{17}$$



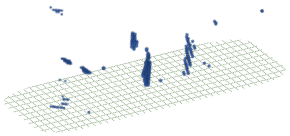
GT	A person walks forward in a straight line and turns back.
MGPT	A person walks backward, with arms swing forward then backswing left hand reaches up and down
VLLaMA3	A person steps in place with alternating weight.
RadarLLM	A person steps forward with a torso roll
Ours	A person steps forward, leans low, and turns.



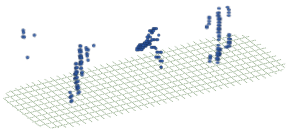
GT	A person does a deep squat and then slowly stands up.
MGPT	A person squats down and stands up repeatedly.
VLLaMA3	A person lunges low and pushes back to standing.
RadarLLM	A person squats low and rises halfway.
Ours	A person squats slightly and extends forward.



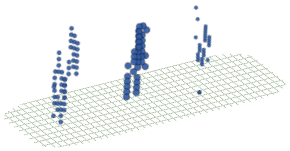
GT	A person takes side steps to the left and to the right with even strides.
MGPT	A person steps back and holds the box.
VLLaMA3	A person sidesteps left with a brief pause.
RadarLLM	A person turns half around and side steps.
Ours	A person leans to one side and steps around the axis.



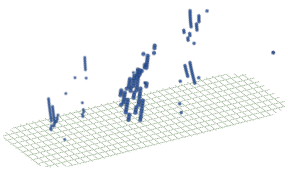
GT	A person kicks a small ball in place.
MGPT	A person side kicks with left leg outward then inward.
VLLaMA3	A person kicks a mat forward and steps after it.
RadarLLM	A person kicks forward and plants the foot.
Ours	A person kicks a ball upward and steps in place.



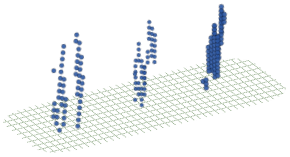
GT	A person sweeps a long broom with the right hand
MGPT	A person extends both arms to the sides and retracts.
VLLaMA3	A person leans to the right and stretches the left arm.
RadarLLM	A person slides one foot in front and shifts down.
Ours	A person sweeps a broom backward and plants the heels.



GT	A person swings a heavy tennis racket and then lowers it.
MGPT	A person alternates steps while stepping side to side.
VLLaMA3	A person turns the sign around and lowers the arm
RadarLLM	A person swings one arm across and resets.
Ours	A person raises a racket overhead and lowers it.



GT	A person lifts a paper cup by hand.
MGPT	A person sets a phone on the table.
VLLaMA3	A person lifts a blanket and lets it fall.
RadarLLM	A person winds up the hips and straightens the spine.
Ours	A person lifts a lamp by the base and lowers it.



GT	A person throws a small ball with both hands.
MGPT	A person dribbles a ball with one hand and changes grip.
VLLaMA3	A person sets a cushion down and smooths the cover.
RadarLLM	A person raises both arms, twists, and lowers them.
Ours	A person throws a toy into the basket and cups both hands.

Figure 8: Additional qualitative results on the real test set. While all methods capture coarse human motions reasonably well, our method better preserves object-related interaction semantics when an object is involved. Fine-grained action details and precise object categories remain challenging.

Thus, $\mathbf{e}_t^{(m)}$ is the multi-head routed observation content for node m. Unlike a normalized weighted average over raw point features, it is computed by applying head-specific routing weights to value-projected point features, concatenating the head outputs, and applying an output projection.

F.2 Routing-Derived Update Gate

In addition to node evidence, we compute a learned per-node update gate $c_t^{(m)} \in [0,1]$ from the same pre-softmax routing logits $\ell_t^{(m,n,h)}$ used to form the assignments. While $\mathbf{e}_t^{(m)}$ provides the routed observation content for node m, $c_t^{(m)}$ controls how strongly this content updates the node relative to the predictive prior.

For each node, we collect three types of routing-logit summary statistics: the per-head maximum logit over valid points, the per-head mean logit over valid points, and the valid-point ratio:

$$\boldsymbol{\rho}_t^{(m)} = \left[\left\{ \max_{n:\eta_t^{(n)}=1} \ell_t^{(m,n,h)} \right\}_{h=1}^{H} ; \left\{ \mathrm{mean}_{n:\eta_t^{(n)}=1} \ell_t^{(m,n,h)} \right\}_{h=1}^{H} ; \frac{1}{N} \sum_{n=1}^{N} \eta_t^{(n)} \right]. \tag{18}$$

The update gate is produced by a small two-layer MLP:

$$c_t^{(m)} = g_\phi\left(\boldsymbol{\rho}_t^{(m)}\right), \qquad g_\phi : \mathbb{R}^{2H+1} \to [0,1]. \tag{19}$$

The sigmoid-bounded output is a routing-derived confidence for the posterior update. It is not a contact label, not an object-interaction score, and not simply the total routing mass of a node.

F.3 Temporal Context and Non-Leakage of the Prior

The predictive prior uses only past latent states. For each past step $i \in \{t-K, \ldots, t-1\}$, we form a context feature by concatenating the posterior node state with its first-order temporal difference $[\mathbf{h}_i; \mathbf{h}_i - \mathbf{h}_{i-1}]$. Each feature is then projected to the context dimension and aggregated by a single-layer GRU:

$$\mathbf{d}_i = f_{\mathrm{ctx}}([\mathbf{h}_i; \mathbf{h}_i - \mathbf{h}_{i-1}]), \qquad \boldsymbol{\xi}_t = \mathrm{GRU}(\mathbf{d}_{t-K}, \ldots, \mathbf{d}_{t-1}), \tag{20}$$

where the final hidden state of the GRU is used as $\boldsymbol{\xi}_t$. At the lower window boundary $i = t-K$, the temporal difference is set to zero. At the start of a sequence ($t < K$), the front of the window is zero-padded in $\mathbf{h}$ (so fewer than K posterior states are available).

The current-frame radar point cloud is not used in computing $\boldsymbol{\xi}_t$. It enters only through the evidence pathway used by the posterior update. Thus, the motion-continuation prior is observation-free with respect to frame t, while the posterior is explicitly observation-conditioned. This separation allows us to interpret the corrective residual $\boldsymbol{\mu}_t^{\mathrm{post}} - \boldsymbol{\mu}_t^{\mathrm{pri}}$ as an observation-driven update rather than a re-encoding of the current frame.

F.4 Context-Conditioned Transition

The transition module expands the shorthand $f_{\mathrm{trans}}(\mathbf{h}_{t-1}, \boldsymbol{\xi}_t)$ used in Eq. (2). The previous posterior node set is first coordinated by inter-node self-attention:

$$\mathbf{H}'_{t-1} = \mathrm{LN}\left(\mathbf{H}_{t-1} + \mathrm{SelfAttn}(\mathrm{LN}(\mathbf{H}_{t-1}))\right), \qquad \mathbf{H}_{t-1} = \{\mathbf{h}_{t-1}^{(m)}\}_{m=1}^{M}. \tag{21}$$

The temporal context $\boldsymbol{\xi}_t$ conditions the transition through FiLM-style feature-wise affine modulation [51], rather than being simply concatenated to the node state. Two linear projections produce node-specific scale and shift parameters:

$$\boldsymbol{\gamma}_t^{(m)} = \mathbf{1} + \alpha \tanh\left(W_\gamma^{(m)} \boldsymbol{\xi}_t\right), \qquad \boldsymbol{\beta}_t^{(m)} = W_\beta^{(m)} \boldsymbol{\xi}_t, \tag{22}$$

where $\alpha = 0.1$ keeps the multiplicative scale close to one for stable training. The FiLM-modulated node representation is

$$\hat{\mathbf{h}}_{t-1}^{(m)} = \boldsymbol{\gamma}_t^{(m)} \odot \mathbf{h}_{t-1}^{\prime(m)} + \boldsymbol{\beta}_t^{(m)}. \tag{23}$$

The transition then advances the node state with a GRUCell, using the previous posterior node state as the recurrent hidden state:

$$\mathbf{h}_{t,\mathrm{tr}}^{(m)} = \mathrm{GRUCell}\left(\hat{\mathbf{h}}_{t-1}^{(m)}, \mathbf{h}_{t-1}^{(m)}\right). \tag{24}$$

We use this transition output as the time-t prior node state,

$$\tilde{\mathbf{h}}_t^{(m)} = \mathbf{h}_{t,\mathrm{tr}}^{(m)}. \tag{25}$$

Finally, the prior distribution parameters are predicted by two linear heads:

$$\boldsymbol{\mu}_t^{\mathrm{pri},(m)} = W_\mu^{\mathrm{pri}} \tilde{\mathbf{h}}_t^{(m)}, \qquad \log \boldsymbol{\sigma}_t^{2,\mathrm{pri},(m)} = \mathrm{clamp}\Big(W_\sigma^{\mathrm{pri}} \tilde{\mathbf{h}}_t^{(m)}, -6, 2\Big). \tag{26}$$

The transition therefore combines inter-node coordination, FiLM-style conditioning by the short-horizon temporal context $\boldsymbol{\xi}_t$, and recurrent state propagation. Since $\boldsymbol{\xi}_t$ is computed only from past latent states (Sec. F.3), the prior remains observation-free with respect to frame t. This preserves the interpretation of $\boldsymbol{\mu}_t^{\mathrm{post}} - \boldsymbol{\mu}_t^{\mathrm{pri}}$ as an observation-driven update from the motion-continuation prior.

F.5 Gated Posterior Update

The posterior module expands the shorthand $f_{\mathrm{post}}(\tilde{\mathbf{h}}_t, \mathbf{e}_t, \mathbf{c}_t)$ used in Eq. (2). For each node, we first fuse the prior node state and the current node evidence to obtain an observation-supported candidate update:

$$\mathbf{h}_{t,\mathrm{fused}}^{(m)} = f_{\mathrm{fuse}}^{\mathrm{node}}\left([\tilde{\mathbf{h}}_t^{(m)}; \mathbf{e}_t^{(m)}]\right). \tag{27}$$

The reliability gate is then used as an interpolation coefficient between this candidate update and the prior node state:

$$\mathbf{g}_t^{(m)} = c_t^{(m)} \mathbf{h}_{t,\mathrm{fused}}^{(m)} + \big(1 - c_t^{(m)}\big) \tilde{\mathbf{h}}_t^{(m)}. \tag{28}$$

Thus, weakly supported or noisy nodes remain closer to the predictive prior, whereas high-confidence nodes can move toward the observation-supported update. Frames with no valid radar points are assigned zero confidence, so unreliable observation evidence is not propagated through the gated update.

After gating, the node set is coordinated by inter-node self-attention:

$$\bar{\mathbf{G}}_t = \mathrm{LN}\left(\mathbf{G}_t + \mathrm{SelfAttn}(\mathbf{G}_t)\right), \qquad \mathbf{G}_t = \{\mathbf{g}_t^{(m)}\}_{m=1}^{M}. \tag{29}$$

Finally, posterior distribution parameters are predicted for each node:

$$\boldsymbol{\mu}_t^{\mathrm{post},(m)} = W_\mu \bar{\mathbf{g}}_t^{(m)}, \qquad \log \boldsymbol{\sigma}_t^{2,\mathrm{post},(m)} = W_\sigma \bar{\mathbf{g}}_t^{(m)}. \tag{30}$$

We do not use an explicit background slot. Instead, invalid points are removed by the input mask, low-confidence nodes are biased toward the prior by the gated update, and downstream soft node selection further reduces the impact of non-informative nodes. The corrective residual $\boldsymbol{\mu}_t^{\mathrm{post}} - \boldsymbol{\mu}_t^{\mathrm{pri}}$ is therefore the observation-driven change from the predictive prior to the confidence-aware, coordinated posterior update. This is the residual signal summarized by the semantic bottleneck for caption generation.

G Dataset Construction

We construct the synthetic radar–caption corpus in four steps: caption generation, text-conditioned motion synthesis, SMPL conversion, and radar rendering. Algorithm 1 summarizes the full pipeline. All synthetic radar clips are rendered from the human body mesh only; no object geometry, scene geometry, walls, or floor are included. Thus, object labels enter the corpus through the interaction caption used to generate the human motion, not through direct object reflections.

Caption generation. Training captions are short English motion descriptions generated with a structured-output language model and filtered before use. Object-involving captions are constrained to a fixed vocabulary $\mathcal{V}$ of everyday objects, and captions are rejected if they contain out-of-vocabulary objects, forbidden scene or intention words, missing action verbs, or near-duplicate phrasing. We mix object-involving captions with object-absent body-motion captions so that the model learns both interaction-centric and locomotion-only descriptions.

Parameter	Value
Radar device	TI IWR6843AOP
Carrier frequency	60.75 GHz
Chirp slope	54.71 MHz/μs
ADC samples / chirp	96
ADC sample rate	2950 kHz
Chirps / frame	96
Frame rate	18 Hz
Range bins	96
Doppler bins	96
Angle bins	64
TX / RX antennas	3 / 4

Table 7: Radar simulation configuration used for synthetic rendering. The configuration follows the TI IWR6843AOP setup used for real-data collection.

Motion synthesis. Each accepted caption is converted into a human motion using MDM as a text-conditioned motion generator. For the HumanML3D setting, MDM generates a 263-dimensional motion representation, which is recovered into a 22-joint skeletal motion sequence consisting of a global root trajectory and local body motion features. We then retarget the recovered skeleton to SMPL body parameters with a neutral body shape $\beta = \mathbf{0}_{10}$ to obtain a posed body mesh for each frame. The resulting mesh sequence is resampled to the 18 fps radar frame grid and used for RF ray tracing.

Radar rendering. Radar signals are rendered with RF-Genesis under a fixed TI IWR6843AOP radar simulation configuration (Table 7). For each motion, we render one clean view and one perturbed view with moderate changes in range, azimuth, elevation, and sensor height. The renderer ray-casts only the SMPL body mesh: objects are never rendered. This design forces the captioning model to infer object-related interactions from body kinematics rather than from explicit object returns.

Point-cloud post-processing. Each rendered FMCW frame is processed with the RF-Genesis CFAR and AoA pipeline. The resulting sensor-frame point cloud is transformed to the world frame, sorted by SNR, cropped to the top 128 points, and zero-padded when fewer points are detected. The final radar clip is stored as $P \in \mathbb{R}^{T_r \times 128 \times 6}$.

Real radar data collection. We collected a real radar evaluation set from consenting adult participants in a controlled indoor setting. Participants were asked to perform predefined object-involving and object-absent motions. The recordings contain only processed mmWave radar point clouds and do not include visual, audio, or personally identifying information.

H Implementation and Reproducibility Details

Architecture. HIRaD uses $M = 10$ latent nodes with node dimension 256 and four attention heads. Each radar frame contains at most 128 points with 4D input features $(x, y, z, v_{\text{doppler}})$. The point encoder is a masked PointNet with hidden dimensions $[64, 128]$ and output dimension 256. We set the temporal context window to $K = 5$ frames, selected from a small validation sweep over $K \in \{3, 5, 7, 10\}$ where performance was stable. The semantic bottleneck dimension is 256. The caption decoder is a frozen Qwen3-8B language model with QLoRA adapters applied to the attention projections. The radar prefix contains one semantic token and M node-level motion tokens.

Training and losses. The model is trained in three stages. First, the predictive latent state and semantic bottleneck are trained with reconstruction, KL, radar–motion contrastive, and semantic alignment losses. Second, the predictive latent state is frozen and the captioning adapters and prefix projections are trained with the autoregressive captioning loss. Third, all trainable modules are jointly finetuned. We use AdamW with cosine scheduling, gradient clipping at 1.0, and bfloat16 mixed

Algorithm 1 Synthetic Radar–Caption Corpus Generation

Require: object vocabulary $\mathcal{V}$; target size N; radar simulation configuration $\mathcal{C}_{\mathrm{radar}}$ (Table 7); MDM text-conditioned motion generator $\mathcal{M}_{\mathrm{mot}}$; RF-Genesis renderer $\mathcal{R}$
Ensure: synthetic corpus $\mathcal{D} = \{(P^{(n)}, c^{(n)})\}_{n=1}^{N}$

$\mathcal{D} \leftarrow \emptyset$
while $|\mathcal{D}| < N$ **do**
  $\{c_b\} \leftarrow \mathrm{GPTBATCH}(\mathrm{rules}, \mathrm{schema})$ ▷ structured caption generation
  for each caption $c \in \{c_b\}$ **do**
    if $\neg\mathrm{PASSFILTERS}(c, \mathcal{V})$ **then**
      continue
    end if
    $k \leftarrow \mathrm{COMPLEXITYBUCKET}(c)$
    $T_m \leftarrow 90 + 30k$ ▷ motion frames at 30 fps
    $\mathbf{h}_{1:T_m} \leftarrow \mathcal{M}_{\mathrm{mot}}(c, T_m)$ ▷ MDM generates a 263-D HumanML3D motion representation
    $X_{1:T_m} \leftarrow \mathrm{RECOVERJOINTS}(\mathbf{h}_{1:T_m})$ ▷ recover 22-joint skeletal motion
    $(\theta^{\mathrm{SMPL}}_{1:T_m}, \mathbf{t}^{\mathrm{SMPL}}_{1:T_m}) \leftarrow \mathrm{RETARGETTOSMPL}(X_{1:T_m})$ ▷ retarget the recovered skeleton to SMPL body pose
    $\beta \leftarrow \mathbf{0}_{10}$ ▷ neutral body shape
    $(\theta^{\mathrm{SMPL}}_{1:T_r}, \mathbf{t}^{\mathrm{SMPL}}_{1:T_r}) \leftarrow \mathrm{RESAMPLE}((\theta^{\mathrm{SMPL}}_{1:T_m}, \mathbf{t}^{\mathrm{SMPL}}_{1:T_m}), 18/30)$ ▷ resample from 30 fps to 18 fps
    for $v \in \{\mathrm{clean}, \mathrm{perturbed}\}$ **do**
      $\pi \leftarrow \mathrm{SAMPLESENSORPOSE}(v)$
      for $r = 1, \ldots, T_r$ **do**
        $V_r \leftarrow \mathrm{SMPL}(\theta^{\mathrm{SMPL}}_r, \beta, \mathbf{t}^{\mathrm{SMPL}}_r)$ ▷ posed body mesh for RF rendering
        $s_r \leftarrow \mathcal{R}.\mathrm{SIGNALTRACE}(V_r, \pi, \mathcal{C}_{\mathrm{radar}})$
        $\widehat{P}_r \leftarrow \mathcal{R}.\mathrm{PROCESSPC}(s_r, \mathcal{C}_{\mathrm{radar}})$ ▷ CFAR + AoA
        $P_r \leftarrow \mathrm{SENSOR2WORLD}(\widehat{P}_r, \pi)$
        $P_r \leftarrow \mathrm{TOPKBYSNRPAD}(P_r, K{=}128)$
      end for
      $P \leftarrow \mathrm{STACK}((P_r)_{r=1}^{T_r}) \in \mathbb{R}^{T_r \times 128 \times 6}$
      $\mathcal{D} \leftarrow \mathcal{D} \cup \{(P, c)\}$
      if $|\mathcal{D}| = N$ **then**
        break
      end if
    end for
  end for
end while
return $\mathcal{D}$

precision. The loss weights are $\lambda_{\mathrm{obs}} = 1.0$, $\lambda_{\mathrm{KL}} = 0.1$, $\lambda_{\mathrm{ctr}} = 1.0$, $\lambda_{\mathrm{sem}} = 0.5$, and $\lambda_{\mathrm{cap}} = 1.0$, with contrastive temperature 0.07. The two contrastive terms also act as a regularizer that has been shown to aid generalization across domains [52], which is relevant to our synthetic-to-real setting. During captioning stages, node tokens in the decoder prefix are randomly dropped with probability $p_{\mathrm{drop}} = 0.1$.

Data and inference. The synthetic corpus contains 10,000 radar–caption clips and is split into train, validation, and synthetic test sets with an 80/10/10 ratio. Real radar clips are used only for evaluation and are not used for training, validation, hyperparameter tuning, or model selection. Radar clips are padded or truncated to 64 frames. Captions are generated deterministically with beam search using 4 beams, repetition penalty 1.2, and no-repeat 3-gram blocking. All reported captioning results are mean $\pm$ standard deviation over three random seeds.

Compute resources. All experiments were run on a single NVIDIA RTX A6000 GPU (48 GB). A full three-stage HIRaD training run takes about 20 GPU-hours, and reported results are averaged over three seeds. Baselines were trained and evaluated on the same GPU under the same protocol. At inference (batch 1, 64-frame clips), clip captioning takes 1.52 s, dominated by LLM decoding, while the radar perception stack takes 217 ms per clip (3.4 ms per frame), within the sensor's 18 Hz budget.

Scaling M to 40 or N to 512 adds under 2% latency, and perception peak memory is about 115 MB of the 6.1 GB total.

Synthetic motion diversity. Measured directly on the generated 3D motions (root-centered 22-joint positions, 484 clips across 13 object-involving and object-absent sub-categories), the mean intra-category pairwise distance is 86% of the inter-category distance, the median within-category nearest-neighbor distance is 53% of the intra-category mean, no near-duplicate motions exist, and at least 95% of captions per category are unique. Diversity is further enforced by near-duplicate caption rejection, stochastic MDM sampling, per-caption length buckets, and two sensor poses per motion.

I Broader Impacts

HIRaD targets privacy-preserving radar-based human activity understanding for assistive sensing, healthcare monitoring, and robotics where cameras are undesirable. Radar sensing can nonetheless be misused for surveillance without consent. We therefore recommend consented data, release of only de-identified radar features and aggregate annotations, and no deployment where users are unaware of or unable to opt out of sensing.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the main contributions and scope of the work: hidden interaction inference from sparse radar, corrective residual trajectories, and language-conditioned caption generation. The claims are supported by the main experiments, ablations, qualitative analysis, and the limitations section, and we avoid claiming direct object detection or contact estimation.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: We include a dedicated Limitations section discussing the ambiguity of residual trajectories, the possibility that large residuals arise from non-interaction motion or radar sparsity, the limited scale of the physical-object validation, the single-person setting, and the restriction to relatively short clips.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper does not present formal theoretical results, theorems, or proofs. The equations define the model architecture, residual representation, and training objective.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The main paper specifies the datasets, train/validation/test splits, baselines, evaluation protocol, and metrics. The appendix provides implementation details, loss weights, training schedule, inference settings, radar simulation configuration, and compute environment.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: Code is released at `https://dlwowlb.github.io/HIRaD`. The synthetic corpus and the real radar benchmark are not publicly released, and the paper and appendix describe their construction in enough detail to regenerate the synthetic data with the released pipeline.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: The main paper specifies the datasets, train/validation/test splits, baselines, evaluation protocol, and metrics. The appendix provides implementation details, loss weights, training schedule, inference settings, radar simulation configuration, and compute environment.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

Justification: Main captioning and interaction results are reported as mean $\pm$ standard deviation over three random seeds. The tables supporting the main claims include these error bars for both synthetic and real benchmarks.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: The appendix reports the GPU type, memory, and approximate wall-clock compute required for a full HIRaD run. All experiments use a single NVIDIA RTX A6000 GPU with 48 GB memory, and one full run takes approximately 20 GPU-hours.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: We reviewed the NeurIPS Code of Ethics and conducted the work in accordance with it. The real radar recordings are used for research evaluation only, and the paper discusses privacy, misuse, and limitations of radar-based human activity sensing.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.

- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Appendix I discusses positive applications of privacy-preserving assistive sensing and robotics, as well as negative risks such as surveillance or non-consensual monitoring. It also describes mitigation practices such as consented data use and de-identified radar features.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The paper does not release a high-risk pretrained generative model, scraped dataset, or personally identifiable sensor dataset. We nevertheless discuss privacy and surveillance risks and recommend consent-based data use and de-identification.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: We cite the original papers for the existing datasets, models, simulators, and baseline methods used in the experiments. The paper does not enumerate the license name and terms for every external asset, and the released code repository lists the licenses of the assets it depends on.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [No]

Justification: The paper introduces a synthetic radar–caption corpus and a real evaluation benchmark and documents their construction in the main paper and appendix. The assets themselves are not publicly released, and only the code is made available.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [Yes]

Justification: The paper uses a small real radar benchmark collected from consenting adult participants and describes the motion protocol at a high level. No crowdsourcing platform was used, no identifying visual or audio data are released, and the benchmark is used only for evaluation.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [No]

    Justification: The real radar data were collected from consenting adult participants in a controlled setting, and no identifying visual or audio data are released. However, we do not report formal IRB or equivalent review approval.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: We use LLMs as part of the research methodology: a structured prompting protocol generates and normalizes motion captions for the synthetic corpus, and a frozen autoregressive language model is used as the caption decoder conditioned on radar-derived prefix tokens. These uses are described in the dataset construction and method sections.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.